theory to organize all geometric knowledge. Lie and Klein clearly articulated the notion of a transformation group, the continuous analog of a permutation group, with promising applications to geometry and differential equations, but they lacked a general theory of the subject. The Erlangen Program implied one aspect of this project — the group classification problem — but Lie had no intention of attacking this bastion at the time. As he later wrote to Klein, "[I]n your essay the problem of determining *all* groups is not posited, probably on the grounds that at the time such a problem seemed to you absurd or impossible, as it did to me" [**22**, pp. 41–42].

By the end of 1873, Lie's pessimism gave way to a much brighter outlook. After dipping into the theory of first order differential equations, developed by Jacobi and his followers, and making considerable advances with his *idée fixe*, Lie finally acquired the mathematical weaponry needed to answer the challenge of the Erlangen Program and to tackle the theory of continuous transformation groups.

Living on the outskirts of Europe, Lie felt quite marginalized in the European mathematics community. No students and very few foreign colleagues were interested in his research. He wrote his papers in German but published them almost exclusively in Norwegian journals, preferring publication speed over wide accessibility. A few years later he learned, however, that one French mathematician had won the *Grand Prix* from the Académie des Sciences for independently obtained results that yielded some special cases of Lie's work on differential equations. Lie realized that his Norwegian publications were not the greatest publicity vehicle, and that he needed to make his work better known in Europe. "If only I could collect together and edit all my results," he wistfully wrote to Klein [**22**, p. 77]. Klein's practical mind quickly found a solution. Klein, who then taught at Leipzig, arranged for the young mathematician Friedrich Engel, a recent doctoral student of his colleague, to go to Christiania and to render Lie a helping mathematical hand.

Lie and Engel met twice daily for a polite conversation about transformation groups. As Engel recalled, Lie carried his theory almost entirely in his head and dictated to Engel an outline of each chapter, "a sort of skeleton, to be clothed by me with flesh and blood"